

Detecting Cloned Voices on a Phone Call

Detectors score **1.23%** on their own benchmark and **37%** on someone else's.

We are building one that closes that gap.

TEAM

[names]

RUNS ON

The callee's phone, during the call

CORPORA

ASVspoof 2019 LA, 2021 LA, In-the-Wild

DATE

September 2026

Ninety seconds of public audio is enough.

- An attacker takes a voice from a podcast or a voicemail.
- A cloning model copies it. **Minutes of work, no GPU farm.**
- The call reaches a family member, an employee, or a bank.
- The victim hears someone they know.

THE QUESTION THIS TALK ANSWERS

A human hears a familiar voice and believes it. **Can a machine tell that it is synthetic?**

Excellent numbers, on **one dataset**.

BEST REPORTED

1.23%

RawGAT-ST, equal error rate on ASVspoof 2019 LA.

BENCHMARK MATURITY

4

ASVspoof challenges since 2015. 54 teams entered the 2021 evaluation.

ARCHITECTURES COMPARED

12

Re-implemented under one protocol. Raw-waveform models win.

THE CATCH

Every number here comes from the same corpus the model trained on. Change the dataset and they fall apart. **That is the problem we are fixing.**

Move it off its own dataset and it **stops working**.

TRAINED ON ASVSPPOOF 2019, THEN TESTED ON:

its own test set

1.23%

another corpus

In-the-Wild, new speakers

37.15%

2026 cloned voices

over a real VoIP call

50.28%

COIN FLIP

same test, retrained

on modern voices

5.81%

the problem is fixable

equal error rate, lower is better

0%

15%

30%

45%

60%

THE GAP IS THE TARGET

A detector at 1% and 37% has a 36-point gap. **Closing that gap is the job.** More ASVspooof data will not do it: 33.9% to 33.1%, because every split comes from the same recordings.

The detector learns the **dataset**, not the voice.

SILENCE ALONE

85%

Accuracy from a classifier given **only the length of the opening silence**.

Müller et al.

ATTACK ON SILENCE ONLY

82.2%

Attack success from changing **only the background noise and the silence**, never the voice.

SiFDetectCracker, ACM MM 2023

WHERE IT LOOKS

0.5–0.6 kHz

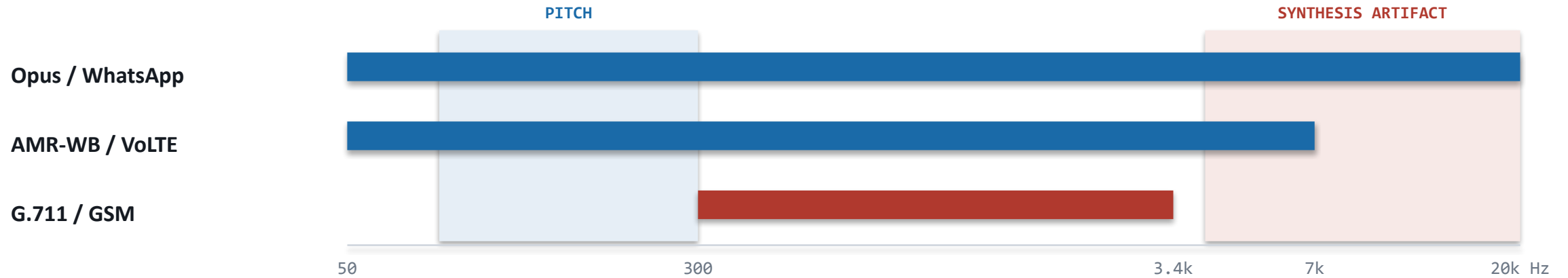
SHAP analysis: the model attends to **the gaps between words** and one narrow low band.

arXiv:2110.03309

WHICH TELLS US WHAT TO FIX

The model wins on things that belong to the corpus, not to the speech. **All of them can be taken away during training.**

What survives a phone line?



GOOD NEWS FOR THE DESIGN

A phone line throws away the top of the spectrum, where the synthesis artifact lives. But low-passing to the telephone band **improves** codec robustness by about 25% relative, because the high bins are what overfit. And the **pitch band alone reaches 1.15% EER**. The band the phone keeps is a band we can build on.

How you build a detector that **travels**.

DESIGN DECISION	WHY
Codec and channel augmentation	Every top-5 system in ASVspoof 2021 used it. No counterexample.
Raw waveform or CQT, never mel	Mel is 37% worse on average, all else equal. It throws away the resolution the task needs.
Silence masking, VAD-trimmed training	Takes away the cue that carries 85% of the decision.
Low-pass to the telephone band	Costs high frequencies, buys 25% relative EER under codec shift. We test it, not assume it.
SSL front-end (wav2vec2)	The most robust under attack: 1.89% to 11.6%, where SSNET goes to 61.81%. Also the heaviest.
One-class objective	Learns what real speech is, instead of which attacks we happened to see.

None of these is new on its own. The work is making them hold against both shifts at once, corpus and channel.

The only number that matters is the **gap**.

TRAIN

ASVspoof 2019 LA, clean, with our augmentation on top. 25,380 utterances.

We hold out one attack family, so unseen attacks are measured before we touch another corpus.

TEST

ASVspoof 2021 LA for the channel. Real VoIP and PSTN, seven codecs, **25,938** trials each.

In-the-Wild for the corpus. New speakers, new attacks, new recordings. 31,779 clips.

OUR HEADLINE METRIC

gap = out-of-domain EER - in-domain EER

A detector at 1% and 37% has a 36-point gap. We report all three, per codec and per corpus, never averaged together. The gap is what we optimise.

An app on the callee's phone.

WHAT THE SETTING GIVES US

A workable budget. A person reads the alarm and needs seconds to react, not 200 milliseconds.

False alarms cost more than misses. An app that cries wolf gets deleted. That sets where we sit on the curve.

WHAT WE MUST NOT HIDE

Android will not hand call audio to a third-party app. VOICE_CALL has needed privileged access since Android 10. iOS is closed.

So we capture through the speaker and the microphone. That adds a room to the signal: **4.7% to 18.2%** EER under replay.

SO WE TRAIN AGAINST IT TOO

Re-capture is a third shift, next to corpus and channel, and it is the one a real deployment forces on us. **It goes in the augmentation stack, not the limitations section.**

One window is not a decision.

THE TRAP

Alarm whenever one window crosses a threshold, at 5% per window and 2-second windows, and a **60-second call false-alarms 79% of the time**. Per-window error is the wrong number to report.

HOW WE DECIDE

Add the evidence up

Accumulate a likelihood ratio across windows and alarm when it crosses a boundary. Wald's sequential test, with separate raise and clear thresholds and a three-second floor.

The decision arrives as soon as the evidence is enough, which is what real-time means here.

WHAT WE REPORT

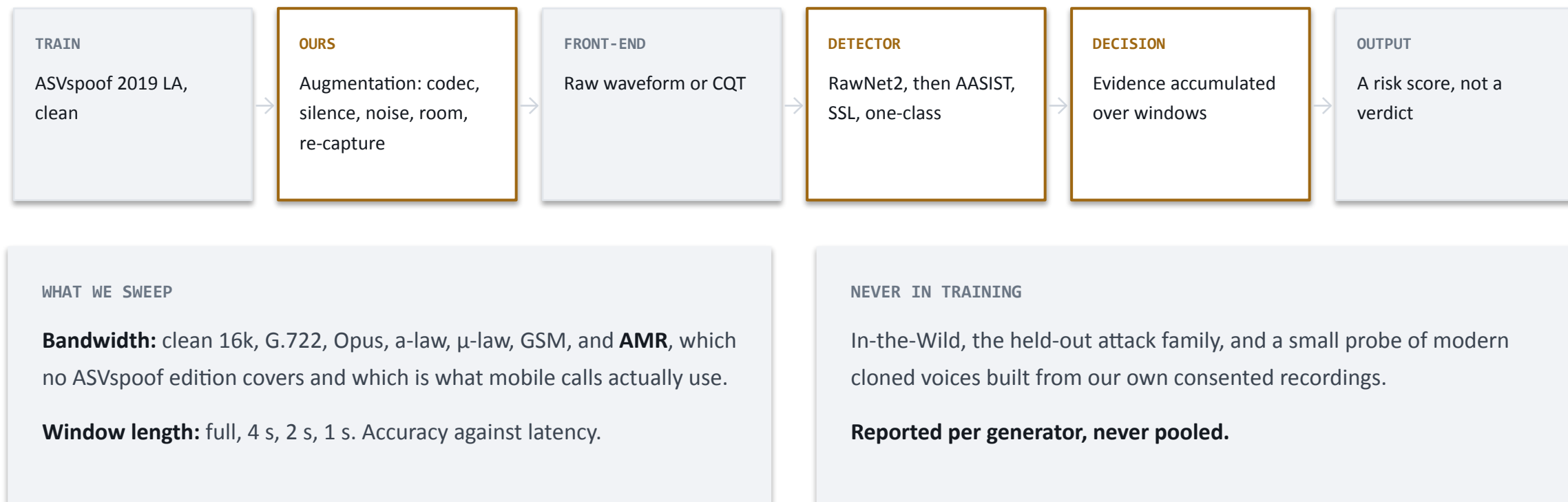
Three numbers, always together

False alarms per call-minute, which is the unit that exposes the trap above.

Detection rate at that operating point.

Time to detection, and milliseconds per window on the phone.

What we will build.



Every codec is already in the ffmpeg build on our machine, and 2021 LA ships the codec label per trial. The sweep costs us no setup.

What success looks like.

THE TARGET

An **out-of-domain EER well below the 34 to 37%** the literature reports on the same held-out data.

A gap we **reduced and can attribute**. Which decision bought how much.

A decision **within seconds of speech**, on the phone, with the cost of every window length measured.

WHAT WE WILL NOT DO

Quote one in-domain number as if it described the field.

Average across corpora or codecs to make a result look better.

Say the problem is solved. A smaller gap is not a closed gap.

THE POSITION WE WILL DEFEND

A detector that travels is worth more than a detector that scores well. **We are optimising the number that survives a change of dataset.**